CHRIST COLLEGE PUNE

DEPARTMENT OF SCIENCE

Programme Outcome

After the successful completion of three-year BSC (CS) Programme, the graduate will be able to:

PO1: To develop problem solving abilities using computer

PO2: To build the necessary skill set and analytical abilities for developing computer-based solutions

PO3: To create awareness about process and product standards

PO4: To train students in professional skills related to software industry

Bachelor of Science (Computer Science)

Semester I

Course Code	Course Title	Course Outcome
CS-111	**Problem Solving using Computer and 'C' Programming**	1. Explore algorithmic approaches to problem solving. 2. Develop modular programs using control structures and arrays in 'C'.
CS-112	**Database Management Systems**	1.Solve real world problems using appropriate set, function, and relational models. 2. Design E-R Model for given requirements and convert the same into database tables. 3. Adapt SQL
MTC-111	**Matrix Algebra**	1. Organize large amount of data in the form of matrices 2. Perform basic matrix operations 3. Solve simple linear equations using matrices
MTC-112	**Discrete Mathematics**	1. Identify certain parameters and properties of the given graphs. 2. Design certain algorithms based on Boolean Algebra
ELC-111	**Semiconductor Devices and Basic Electronic Systems**	1. Define and explain electronics components 2. Relate working of semiconductor devices 3. Develop working of Power supply

		4. Examine working of OSCILLATORS 5. Choose appropriate and advanced techniques for data conversion
ELC-112	**Principles of Digital Electronics**	1. Assess interconversion of various number systems. 2. Design and simplify K-maps 3. Evaluate and design encoders and decoders.
CSST-112	**Mathematical Statistics**	1. Solve counting problems related to permutations and combinations using counting principles 2. Assess probability theory in modelling and understanding modern definition of probability. 3. Identify various probability models. 4. Discover numerical problems related to real life situation
CSST-111	**Descriptive Statistics**	1. Learn to describe and summarize data using appropriate measures of central tendency, such as mean, median, and mode. 2. Understand measures of data variability, including range, variance, and standard deviation, to assess the spread of data points. 3. Calculate and interpret percentiles and quartiles to understand data distribution and identify outliers.

Semester II

Course Code	Course Title	Course Outcome
CS-121	**Advanced 'C' Programming**	1. Recall the basic concepts of 'C' Programming 2. Understand code organization with complex data types and structures.

		3. Develop modular programs using control structures, pointers, arrays, strings and structures. 4. Differentiate various types of outputs.
CS-122	**Relational Database Management Systems**	1. Classify and explain the database concepts like data types, operators and control statements. 2. Apply the database concepts in writing PL/SQL programs. 3. Distinguish between DBMS and RDBMS 4. Select from various programming constructs like cursors, triggers, functions and procedures and use them while creating programs. 5. Create and organise data in tables and make modifications through PL/SQL programs
MTC-121	**Linear Algebra**	1. Recall the concept of Vector Spaces . 2. Compute basis of given vector spaces. 3. Solve the system of linear equations using the gaussian elimination method. 4. Represent different isomorphisms of vector spaces using linear transformation. 5. Apply Rank-Nullity Theorem for computations of dimension of vector spaces
		1. Understand the concept of Vector Spaces. 2. Compute basis of given vector spaces.

MTC-122	**Graph Theory**	3. Solve the system of linear equations using the gaussian elimination method. 4. Represent different isomorphisms of vector spaces using linear transformation. 5. Apply Rank-Nullity Theorem for computations of dimension of vector spaces.
ELC-122	**Basics of Computer Organisation**	1. Expalin the working of sequential circuits. 2. Compare and identify the basics of computers. 3. Develop the understanding and identifying the memory architecture. 4. Design their own model using basic knowledge of digital electronic
CSST-121	**Methods Of Applied Statistics**	1. Explain the basic concepts of applied statistics. 2. Analyse the data using correlation and regression 3. Demonstrate the method for solving real life problems using multiple regression and multiple, partial correlation. 4. Formulate real life problems as relevant statistical problems and find the solution using methods of time series

CSST-122	**Continuous Probability Distributions and Testing of Hypothesis**	1. Understand standard continuous probability distributions. 2. Compute probabilities, expected values and variance of standard continuous probability distributions. 3. Apply standard continuous probability distributions to solve real life problems based on given data. 4. Implement testing of hypothesis in decision making situation. 5. Create optimum model against which given data needs to be tested.
ELC-121	**Instrumentation System**	1. Infer the fundamental principles of measurement, including accuracy, precision, sensitivity, and linearity. 2. Familiarize yourself with common measurement instruments, such as voltmeters, ammeters, and oscilloscopes 3. Explore digital communication protocols used in instrumentation systems, such as Modbus and HART.

Semester III

Course Code	Course Title	Course Outcome
CS-231	**Data Structures and Algorithms – I**	1. Interpret organized data structures in PS. 2. Differentiate the usage of various structure in PS. 3. Implementing algorithms to solve problems using DS.
CS-232	**Software Engineering**	1. Compare and chose a process model for a software project development. 2. Identify requirements analyze and prepare models. 3. Prepare the SRS, Design document, Project plan of a given software system

MTC-231	**Group Theory**	1. Differentiate between equivalence relation and congruence relation on Z 2. Identify groups and its elementary properties 3. Explore the properties of finite groups and subgroups
MTC-232	**Numerical Analysis**	1. Develop methods like Newton-Raphson, false position while solving algebraic and transcendental equations. 2. Apply methods like Newton's interpolation, LaGrange's interpolation while solving interpolation problem. 3 Implement methods like Trapezoidal Rule, Simpson's One-Third Rule, Simpson's Three-Eight Rule while finding numerical values of integration 4. Examine methods like Euler's Method, Modified Euler's Method, Runge-Kutta Method while solving ordinary differential equations
ELC-231	**Microcontroller Architecture and programming**	1. Classify programs for 8051 microcontroller 2. Interface I/O peripherals to 8051 microcontroller 3. Design small microcontroller based projects
ELC-232	**Digital Communication and Networking**	1. Define and explain terminologies of data communication 2. Understand the impact and limitations of various digital modulation techniques 3. Acknowledge the need of spread spectrum schemes 4. Identify functions of data link layer and network layer while accessing communication link 5. Choose appropriate and advanced techniques to build the computer network
23921	**Environment Science –I**	1. Expalin the importance of preserving and protecting our Mother Earth. 2. Guide the next generation.

23922	**Language Communication – I**	1. Develop Reading, Writing, Speaking and Listening Skills. 2. Compare certain rules of grammar 3. Develop personality by instilling core values; life skills. 4. Identify soft skills.

Semester IV

Course Code	Course Title	Course Outcome
CS-242	**Computer Networks - I**	1. Recall the basic concept of Networking(LAN,MAN,WAN, Protocols) 2. Develop the OSI and TCP/IP Reference Models and in particular have a good knowledge of lower Layers vs Upper layers and also working of various protocols. 3. Analyze the requirements for a given organizational structure and select the most appropriate networking architecture and technologies.
		1. Understand 2 and 3 dimensional transformations. 2. Compute matrices of two and three dimensional transformations. 3. Apply transformations and projections in computer science.

MTC-241	**Computational Geometry**	4. Represent different equations of planes and curves in three dimensions. 5. Create animation videos using transformations and curves in three dimensions.
MTC-242	**Operations Research**	1. Explain the basic concepts of operations research. 2. Solve LPP using various methods. 3. Demonstrate the method for solving assignment problems and transportation problems. 4. Formulate real life problems as relevant LPP and find the solution using appropriate method.
ELC-241	**Embedded System Design**	1. Differentiate between general computing and embedded systems. 2. Relate the fundamentals of embedded systems. 3. Develop familiarity with tools used to develop in an embedded environment 4. Design their own model using a Single board Computer (Such as Raspberry Pi) for an embedded system application. 5. Design embedded systems and their interfaces with peripheral devices.

ELC-242	**Wireless Communication and Internet of Things**	1. Understand working of wireless technologies such as Mobile communication, GSM, GPRS ,3G and 4G Cellular Network Technologies 2. Apply short-range communication in various applications 3. Experiment above knowledge for data communication 4. Evaluate the concept of Internet of Things 5. Design new IoT based application
24921AECC1	**Environment Science – II**	1. Make use of awareness in the society about the protection of the environment. 2. Create awareness about natural resources, ecosystem, biodiversity and its conservation in urban, rural, industrial, agricultural areas. 3. Discuss issues related to society, human population, environmental pollution etc. 4. Investigate scope of renewable energy
CS-241	**Data Structures & Algorithms - II**	1. Learn about advanced data structures such as heaps, hash tables, balanced trees (e.g., AVL trees, Red-Black trees), and graphs. 2. Understand the principles of dynamic programming and apply it to solve complex problems, including optimization problems and sequence alignment.
24922	**Language Communication II**	1. Develop a higher level of proficiency in the target language, including improved speaking, listening, reading, and writing skills. 2. Achieve greater fluency in spoken language through practice in real-life situations and discussions.

		3. Enhance reading comprehension skills to understand more complex texts and literature.

Semester V

Course Code	Course Title	Course Outcome
CS-351	**Operating Systems - I**	1. Assess concept of operating system and its principle 2. Analyze processes and Thread Scheduling by operating system 3. Create synchronization in process and threads by operating system 4. Design memory management by operating system using with the help of various schemes
CS-352	**Computer Networks – II**	1. Understand the different protocols of Application layer. 2. Develop understanding of technical aspect of Multimedia Systems. 3. Relate various Multimedia Systems applicable in real time. 4. Identify information security goals. 5. Compare and apply cryptographic techniques for data security
CS-353	**Web Technologies – I**	1. How to develop dynamic and interactive Web Page 2. Create well-structured web pages using HTML. 3. Implement user authentication and authorization mechanisms
		1. Perform Exploratory Data Analysis

CS-354	**Foundations of Data Science**	2. Obtain, clean/process, and transform data. 3. Detect and diagnose common data issues, such as missing values, special values, outliers, inconsistencies, and localization. 4. Demonstrate proficiency with statistical analysis of data. 5. Present results using data visualization techniques.
CS-355	**Object Oriented Programming - I (Core Java)**	1. Understand the concept of Object-Oriented Programming such as classes, objects, Packages, and Collections. 2. Develop Web-based applications and GUI-based Applications. 3. Design event-driven Applications.
CS-356	**Theoretical Computer Science and Compiler Construction - I**	1. Recall the Finite Automata, Pushdown Automata and Turing Machine. 2. Evaluate the Regular Language, Context Free Language, Context Sensitive Language and Unrestricted Language. 3. Explain the relation between Automaton and Language
CS-3510	**Python Programming**	1. Develop logic of problem-solving using python 2. Evaluate various Data Structures like List 3. Familiarize with Basic control structures, loops & tune? 4. Develop small Application Project.
CS-3511	**Blockchain Technology**	1. Learn Fundamentals of Blockchain Technology. 2. Learn Blockchain Programming. 3. Demonstrate basic knowledge of smart contracts & how they function

Semester VI

Course Code	Course Title	Course Outcome
CS-361	**Operating Systems – II**	1. Compare deadlock concepts and file handling techniques. 2. Apply disk scheduling algorithms in process management 3. Analyse the working of distributed and mobile O.S.
CS-362	**Software Testing**	1. Define various software testing methods and strategies. 2. Compare a variety of software metrics and latest testing methods used in the software industries. 3. Identify defects and manage those defects for improvement in quality for given software. 4. Design test cases and test plans, review reports of testing for qualitative software.
CS-363	**Web Technologies – II**	1. Recall the basic concepts of HTML, CSS, PHP 2. Remember basic concepts of JavaScript, XML, jQuery, cookies, session, Ajax, CodeIgniter 3. Analyse the difference between server-side Application and client-side Application

		4. Create dynamic website. Using MVC based framework, students will make design and handle the errors in dynamic Website
CS-364	**Data Analytics**	1. Utilize knowledge and skills to continue learning and adapting to new data science technologies. 2. Acquire the fundamental understanding of the analytical techniques and software tools necessary to effectively generate useful information from structured and unstructured datasets of any size. 3. Demonstrates the ability to technical skills in predictive and prescriptive modelling to support decision making based on data analytics. 4. Evaluate to the basics of programming and analytics software used to the solution of real-world business problems.
CS-365	**Object Oriented Programming using Java – II**	1. Recall basic concepts of Database programming, Collection Framework, Multithreading, and dynamic web pages using Servlets and JSP and also basics Spring Framework. 2. Analyze the difference between Java Application and Web Application 3. Create application using Multithreading concepts, database application using JDBC and web application using Servlets and Spring Framework.

CS-3610	**Software Testing Tools**	1. Define various software testing methods and strategies. 2. Evaluate a variety of software metrics and latest testing methods used in the software industries. 3. Identify defects and manage those defects for improvement in quality for given software. 4. Design test cases and test plans, review reports of testing for qualitative software
CS-366	**Compiler Construction**	1. Remember of the overall structure and components of a compiler, including lexical analysis, parsing, semantic analysis, optimization, and code generation. 2. Explore semantic analysis to check and enforce the correctness of program semantics, including type checking and error handling. 3. Develop techniques for managing symbol tables, which store information about variables, functions, and other program entities. 4. Design and implement test cases and techniques for thoroughly testing the compiler to ensure its correctness and robustness.